



CLASSROOM CONTACT PROGRAMME

(Academic Session : 2022 - 2023)

JEE(Advanced)
UNIT TEST
13-02-2022

JEE(Main + Advanced) : ENTHUSIAST COURSE [PHASE : TEAS]

Time : 3 Hours

Maximum Marks : 198

READ THE INSTRUCTIONS CAREFULLY

GENERAL :

1. This sealed booklet is your Question Paper. Do not break the seal till you are told to do so.
2. Use the Optical Response sheet (ORS) provided separately for answering the questions.
3. Blank spaces are provided within this booklet for rough work.
4. Write your name, form number and sign in the space provided on the back cover of this booklet.
5. After breaking the seal of the booklet, verify that the booklet contains **36** pages and that all the **18** questions in each subject and along with the options are legible. If not, contact the invigilator for replacement of the booklet.
6. You are allowed to take away the Question Paper at the end of the examination.

OPTICAL RESPONSE SHEET :

7. The ORS will be collected by the invigilator at the end of the examination.
8. Do not tamper with or mutilate the ORS. **Do not use the ORS for rough work.**
9. Write your name, form number and sign with pen in the space provided for this purpose on the ORS. **Do not write any of these details anywhere else on the ORS.** Darken the appropriate bubble under each digit of your form number.

DARKENING THE BUBBLES ON THE ORS :

10. Use a **BLACK BALL POINT PEN** to darken the bubbles on the ORS.
11. Darken the bubble  **COMPLETELY.**
12. The correct way of darkening a bubble is as : 
13. The ORS is machine-gradable. Ensure that the bubbles are darkened in the correct way.
14. Darken the bubbles **ONLY IF** you are sure of the answer. There is **NO WAY** to erase or "un-darken" a darkened bubble.
15. Take **$g = 10 \text{ m/s}^2$** unless otherwise stated.

QUESTION PAPER FORMAT :

16. The question paper has three parts : Physics, Chemistry and Mathematics.

Please see the last page of this booklet for rest of the instructions

DO NOT BREAK THE SEALS WITHOUT BEING INSTRUCTED TO DO SO BY THE INVIGILATOR

SOME USEFUL CONSTANTS

Atomic No. : H = 1, B = 5, C = 6, N = 7, O = 8, F = 9, Al = 13, P = 15, S = 16, Cl = 17, Br = 35, Xe = 54, Ce = 58

Atomic masses : H = 1, Li = 7, B = 11, C = 12, N = 14, O = 16, F = 19, Na = 23, Mg = 24, Al = 27, P = 31, S = 32, Cl = 35.5, Ca = 40, Fe = 56, Br = 80, I = 127, Xe = 131, Ba = 137, Ce = 140

- **Boltzmann constant** $k = 1.38 \times 10^{-23} \text{ JK}^{-1}$
- **Coulomb's law constant** $\frac{1}{4\pi\epsilon_0} = 9 \times 10^9$
- **Universal gravitational constant** $G = 6.67259 \times 10^{-11} \text{ N-m}^2 \text{ kg}^{-2}$
- **Speed of light in vacuum** $c = 3 \times 10^8 \text{ ms}^{-1}$
- **Stefan-Boltzmann constant** $\sigma = 5.67 \times 10^{-8} \text{ Wm}^{-2}\text{-K}^{-4}$
- **Wien's displacement law constant** $b = 2.89 \times 10^{-3} \text{ m-K}$
- **Permeability of vacuum** $\mu_0 = 4\pi \times 10^{-7} \text{ NA}^{-2}$
- **Permittivity of vacuum** $\epsilon_0 = \frac{1}{\mu_0 c^2}$
- **Planck constant** $h = 6.63 \times 10^{-34} \text{ J-s}$

Space for Rough Work

HAVE CONTROL → HAVE PATIENCE → HAVE CONFIDENCE ⇒ 100% SUCCESS

BEWARE OF NEGATIVE MARKING

PART-1 : PHYSICS

SECTION-I (i) : (Maximum Marks: 18)

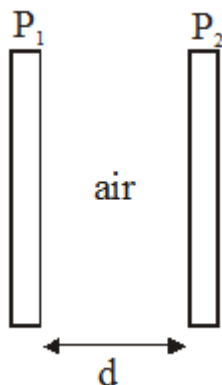
- This section contains **SIX (06)** questions.
- Each question has Four options (A), (B), (C) and (D). **ONLY ONE** of these four options is correct.
- For each question, choose the option corresponding to the correct answer.
- Answer to each question will be evaluated according to the following marking scheme :

Full Marks : +3 If **ONLY** the correct option is chosen.

Zero Marks : 0 If none of the options is chosen (i.e. the question is unanswered)

Negative Marks : -1 In all other cases

1. Two conducting large plates P_1 and P_2 are placed parallel to each other at very small separation 'd'. The plate area of either face of plate is A. A charge $+2Q$ is given to plate P_1 and $-Q$ to the plate P_2 (neglect ends effects). If plate P_1 and P_2 are now connected by conducting wire, then total amount of heat produced is:



(A) $\frac{4Q^2d}{3\epsilon_0 A}$

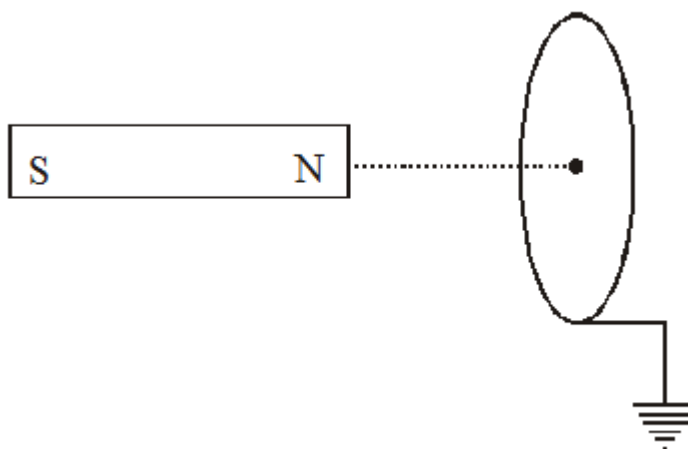
(B) $\frac{9Q^2d}{8\epsilon_0 A}$

(C) $\frac{3Q^2d}{4\epsilon_0 A}$

(D) None of these

Space for Rough Work

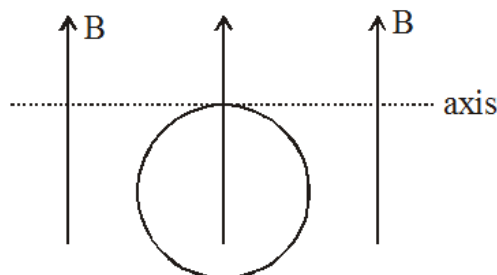
2. A circular ring is fixed in a gravity free space and one point of the ring is earthed. Now a magnet is placed along axis of the ring at a distance from its centre such that the nearer pole is north pole as shown in figure. A sharp impulse is applied on the magnet so that it starts to move towards the ring. Then.



- (A) Initially magnet experiences an acceleration and then it retards to come to an instantaneous rest.
- (B) Magnet starts to oscillate about centre of the ring
- (C) Magnet continuous to move along the axis with constant velocity
- (D) The magnet retards and comes to rest finally.

Space for Rough Work

3. A metallic ring of radius a , mass m and resistance R is free to rotate about a horizontal axis passing through its periphery. The axis is parallel to its horizontal diameter as shown. A uniform magnetic field is present in vertically upward direction. When the ring is pulled by a small angle and released, it executes oscillations. What is the angular frequency of its oscillations?

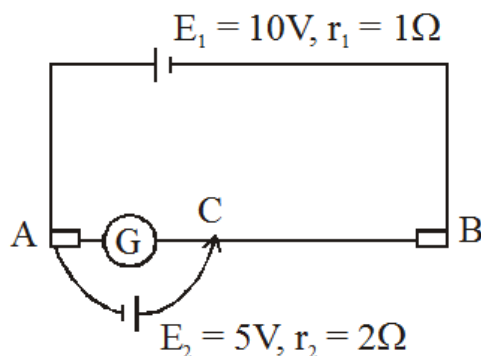


- (A) $\sqrt{\frac{2g}{3a} - \left(\frac{\pi^2 a^2 B^2}{mR}\right)^2}$ (B) $\sqrt{\frac{2g}{3a} - \left(\frac{\pi^2 a^2 B^2}{2mR}\right)^2}$
 (C) $\sqrt{\frac{2g}{3a} - \left(\frac{\pi^2 a^2 B^2}{3mR}\right)^2}$ (D) $\sqrt{\frac{2g}{3a} - \left(\frac{\pi^2 a^2 B^2}{4mR}\right)^2}$
4. An LC circuit consists of a coil of inductance L with negligible resistance and parallel plate capacitor of capacity C , the distance between the plates is d . The capacitor plates are not charged, and there is no current in the circuit. During a time $\Delta t \ll \frac{1}{\sqrt{LC}}$ we gave a charge to the capacitor so that there is an electric field E directed perpendicular to the plates. The coil remained out of the electric field. What will be the maximum current in the circuit thereafter ?

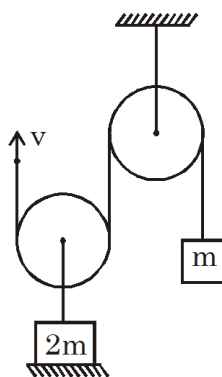
- (A) $I = 2Ed\sqrt{\frac{C}{L}}$ (B) $I = Ed\sqrt{\frac{C}{2L}}$ (C) $I = Ed\sqrt{\frac{2C}{L}}$ (D) $I = Ed\sqrt{\frac{C}{L}}$

Space for Rough Work

5. For the potentiometer arrangement shown in the figure, length of wire AB is 100 cm and its resistance is 9Ω . Find the length AC for which the galvanometer G will show zero deflection.



- (A) 66.7 cm (B) 60 cm (C) 50 cm (D) 33.3 cm
6. In the system shown in figure, strings are weightless and inextensible, the pulleys are weightless, there is no friction. At first, the thread on the left is held so that the load of mass m hangs motionless, and the load of mass $2m$ touches floor. A person holding the end of the thread begins to pull it upwards with a constant speed v . How will it move both the loads ? (v_1 = velocity of m , v_2 = velocity of $2m$)



- (A) $v_1 = v_2 = v/2$ in the upwards direction.
- (B) $v_1 = v/3$ and $v_2 = 2v/3$ both in the upwards direction.
- (C) $v_1 = v_2 = v/3$ both in the upwards direction.
- (D) $v_1 = v/2$ and $v_2 = v/4$ both in the upwards direction.

Space for Rough Work

SECTION-I (ii) : (Maximum Marks: 24)

- This section contains **SIX (06)** questions.
- Each question has **FOUR** options. **ONE OR MORE THAN ONE** of these four option(s) is (are) correct answer(s).
- For each question, choose the option(s) corresponding to (all) the correct answer(s)
- Answer to each question will be evaluated according to the following marking scheme:

Full Marks : +4 If only (all) the correct option(s) is (are) chosen.

Partial Marks : +3 If all the four options are correct but **ONLY** three options are chosen.

Partial Marks : +2 If three or more options are correct but **ONLY** two options are chosen and both of which are correct.

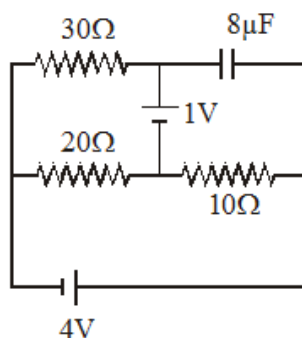
Partial Marks : +1 If two or more options are correct but **ONLY** one option is chosen and it is a correct option.

Zero Marks : 0 If none of the options is chosen (i.e. the question is unanswered).

Negative Marks : -2 In all other cases.

- **For Example** : If first, third and fourth are the **ONLY** three correct options for a question with second option being an incorrect option; selecting only all the three correct options will result in +4 marks. Selecting only two of the three correct options (e.g. the first and fourth options), without selecting any incorrect option (second option in this case), will result in +2 marks. Selecting only one of the three correct options (either first or third or fourth option), without selecting any incorrect option (second option in this case), will result in +1 marks. Selecting any incorrect option(s) (second option in this case), with or without selection of any correct option(s) will result in -2 marks.

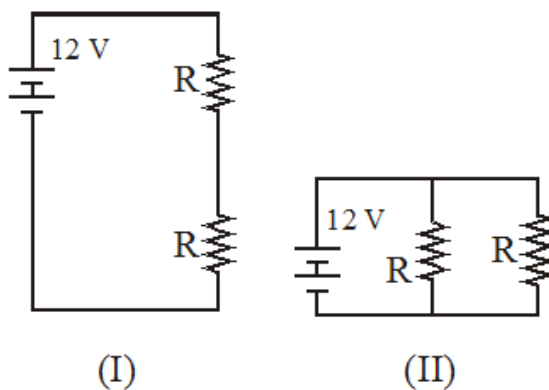
7. For the circuit shown in the figure, in the steady state,



- (A) The current through 30 Ω resistor is 0.2 A. (B) The current through cell of 1 V is 0.1 A.
 (C) The current through cell of 4 V is 0.2 A. (D) The charge on the 8 μ F capacitor is 8 μ C.

Space for Rough Work

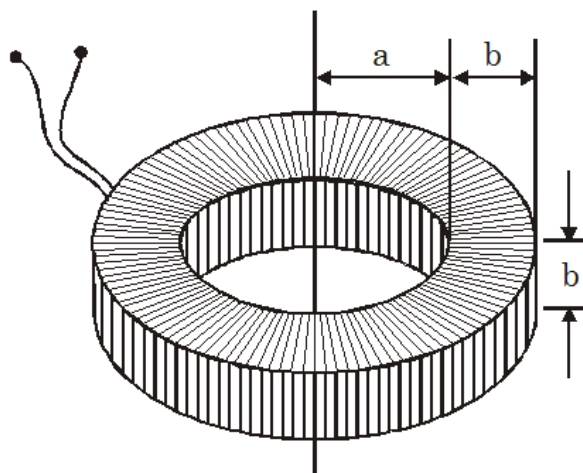
8. Two identical resistors with resistance R are connected in the two circuits drawn. The battery in both circuits is a 12 volt ideal battery. Which statement is/are correct?



- (A) More current will flow through each R in circuit I than in circuit II.
 (B) More total power will be delivered by the battery in circuit II than in circuit I.
 (C) The potential drop across each R will be greater in circuit I than in circuit II.
 (D) The power dissipated in each R will be greater in circuit II than in circuit I.

Space for Rough Work

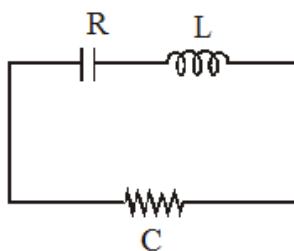
9. Consider a toroid of square cross section and a long straight wire on its axis as shown in the figure. The total number of turns in the toroid is N .



- (A) The mutual inductance of the system is $\frac{\mu_0 N b}{2\pi} \ln \left(\frac{a+b}{a} \right)$
- (B) If the straight wire is displaced parallel to itself so that it remains inside the toroid but not on the axis, mutual inductance decreases.
- (C) If the straight wire is displaced parallel to itself so that it remains inside the toroid but not on the axis, mutual inductance increases.
- (D) If the straight wire is displaced parallel to itself so that it remains inside the toroid but not on the axis, mutual inductance remains same.

Space for Rough Work

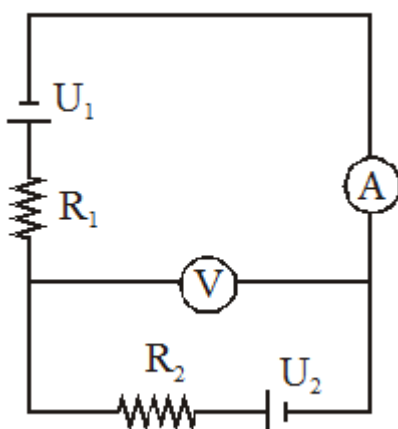
10. Two identical coils of negligible resistance, when connected in series across a 50 Hz fixed voltage source of voltage 314 V, draw a current of 10 A. When the terminals of one of the coils are reversed, the current drawn is 8A.
- (A) The total inductance in 1st situation is more than that in the second case.
- (B) In both cases, current lag's behind the applied voltage
- (C) Mutual inductance is 6.25×10^{-3} H.
- (D) Coefficient of coupling is $\frac{1}{9}$.
11. In a series LCR circuits, charge on the capacitor is varying according to the equation.
 $Q = (4\mu\text{C})e^{-0.2t} \sin(4\pi t + \pi/3)$ (t in sec).



- (A) Current in the resistor is zero at $t = \frac{1}{6}$ sec.
- (B) The emf across the inductor is zero at $t = \frac{2}{3}$ sec
- (C) The charge on the capacitor is zero at $t = \frac{1}{6}$ sec
- (D) The time interval between two successive times when charge on capacitor zero is 0.5 sec.

Space for Rough Work

12. The terminal voltages of the power supplies shown in the figure are constant, and the values of the terminal voltage are: $U_1 = 15 \text{ V}$ and $U_2 = 45 \text{ V}$. The resistances of the resistors are $R_1 = 10\Omega$ and $R_2 = 20\Omega$. The meters are ideal.



- (A) In the figure shown, the reading of the voltmeter is 35 V.
 (B) In the figure shown, the reading of the ammeter is 1 A.
 (C) If the position of voltmeter and ammeter is exchanged, the new reading of the ammeter is 2.25 A.
 (D) If the position of voltmeter and ammeter is exchanged, the new reading of the Voltmeter is 45 V.

Space for Rough Work

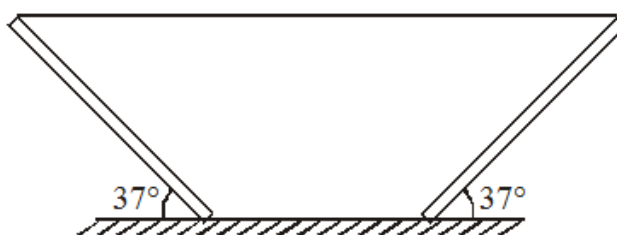
SECTION-II : (Maximum Marks: 24)

- This section contains **SIX (06)** questions. The answer to each question is a **NUMERICAL VALUE**.
- For each question, enter the correct numerical value of the answer in the place designated to enter the answer. If the numerical value has more than two decimal places, **truncate/round-off** the value to **Two** decimal places; e.g. 6.25, 7.00, -0.33, -.30, 30.27, -127.30, if answer is 11.36777..... then both 11.36 and 11.37 will be correct)
- Answer to each question will be evaluated according to the following marking scheme:

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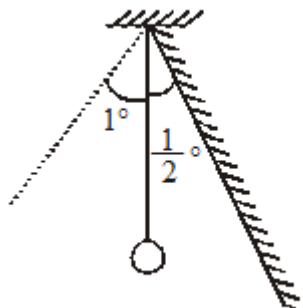
Zero Marks : 0 In all other cases.

1. Two uniform identical rods of same mass are tied together with the help of a string and balanced as shown in the figure. The minimum coefficient of friction for which the system will remain in equilibrium in the position is:



Space for Rough Work

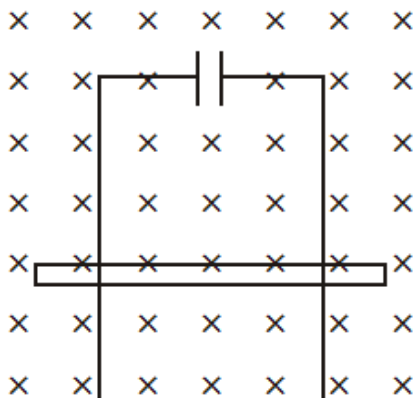
2. A small pendulum bob is connected to 1m long thread. The bob has a mass of 1kg and the bob experiences an air resistance of $\vec{F} = -b\vec{v}$ where $b = 2 \text{ kg/sec}$. Then bob is pulled to left side by 1° and released. The collision of bob with wall as shown is completely elastic. How many times can be bob collide with the wall ?



3. Disc of mass $M = 10 \text{ kg}$ and radius $R = 50 \text{ cm}$ is revolving about fixed vertical axis on smooth horizontal surface with angular velocity $\omega_0 = 4 \text{ rad/sec}$. A block of mass $m = 5 \text{ kg}$ is connected to disc through slack string as shown in the diagram. Find the velocity (in m/s) of the block when string become taut.

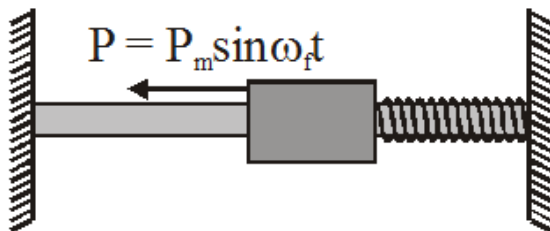
Space for Rough Work

4. A capacitor of capacitance $100 \mu\text{F}$ is connected to smooth railing and connector of mass 1 kg and length 1 m , whole system is in vertical plane and a magnetic field of strength 100 T is applied perpendicular to plane of circuit as shown. Connector is released under gravity. The time (in sec) after which dielectric of capacitor will burn out is $k \times 10^3$ then find the value of k . Breaking strength of dielectric is $5 \times 10^8 \text{ V/m}$. (d is the distance between plates of capacitor = 1 mm)

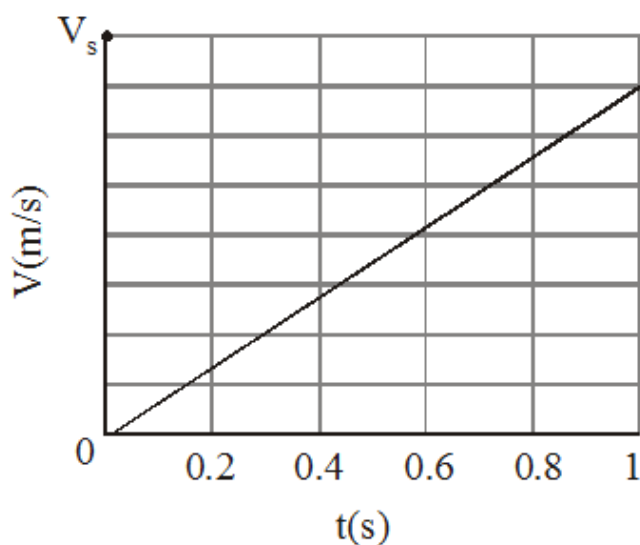


Space for Rough Work

5. A 4 kg slider which slides on a frictionless horizontal rod is attached to a spring of constant k and is acted upon by a periodic force of $P = P_m \sin \omega_f t$. $P_m = 9\text{ N}$ and $\omega_f = 5\text{ rad/s}$. What is the value of the spring constant (in N/m) if the slider has an amplitude of 0.15 m and its motion is out of phase with the applied force?



6. Figure gives the speed v versus time t for a 0.500 kg object of radius 6.00 cm that rolls smoothly down a 30° ramp. The scale on the velocity axis is set by $V_s = 4.0\text{ m/s}$. What is the rotational inertia of the object? Given answer in kg cm^2 .



Space for Rough Work

PART-2 : CHEMISTRY
SECTION-I (i) : (Maximum Marks: 18)

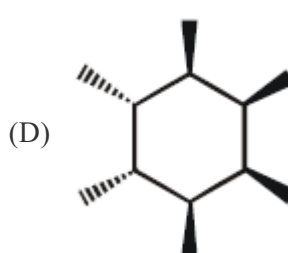
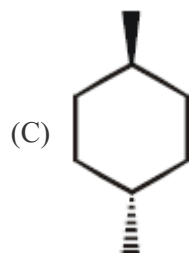
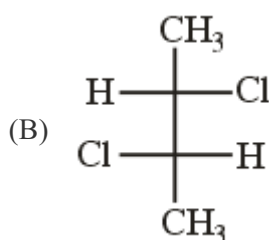
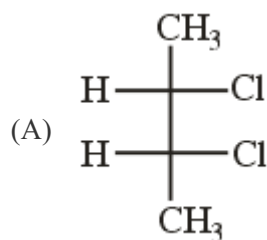
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Negative Marks : -1 In all other cases

1. Which of the following compound is optically active with C_2 -axis of symmetry ?

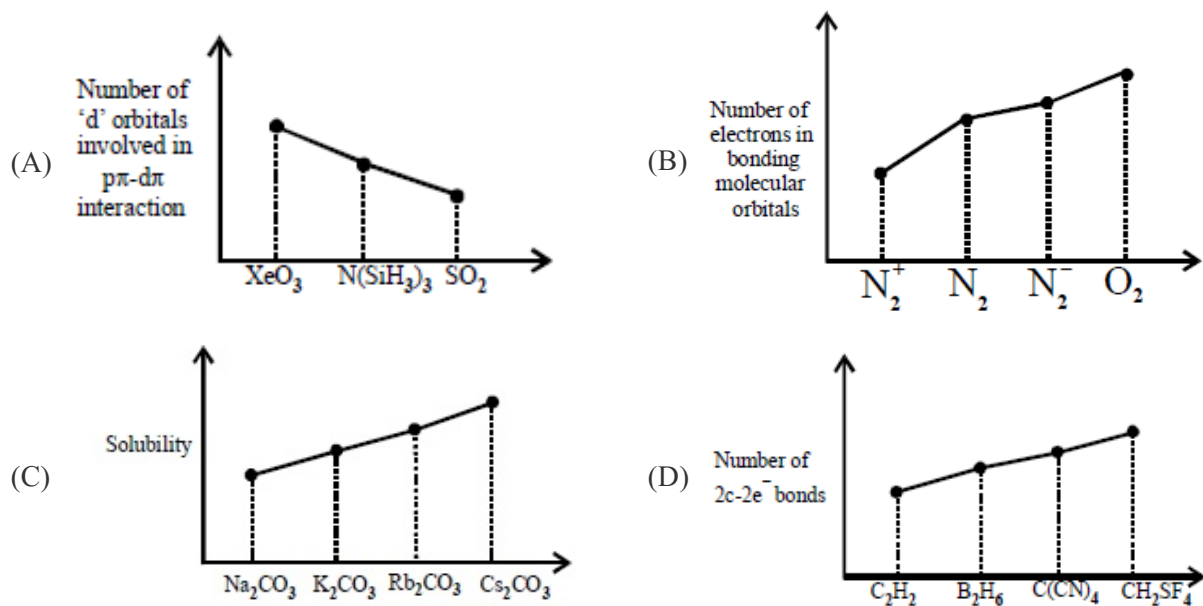


Space for Rough Work

2. Select correct statement(s) among the following
- (A) Polar protic solvent favours S_N2 mechanism more than polar aprotic solvent generally
- (B) Nucleophilic order of halides in polar protic solvent is $I^- > Br^- > Cl^- > F^-$
- (C) Always perfect racemisation take place in S_N1 mechanism
- (D) Inversion of configuration in S_N2 mechanism always produces R to S configuration or S to R configuration
3. On mixing 15.0 ml of ethyl alcohol of density 0.792 g ml^{-1} with 15 ml of pure water at 4°C , the resulting solution is found to have a density of 0.924 g ml^{-1} . The percentage contraction in volume is :
- (A) 8 % (B) 2 % (C) 3 % (D) 4 %
4. 1 litre of an aqueous solution of urea having density = 1.060 gram/mL is found to have $\Delta T_b = 0.5^\circ\text{C}$. If temperature of this solution is increased to 101.5°C then calculate amount of water which must have gone in vapour state upto this point.
Given $K_b = 0.5 \text{ K kg mol}^{-1}$ of water.
- (A) 666.67 gram (B) 566.67 gram (C) 466.67 gram (D) 366.67 gram

Space for Rough Work

5. Which of the following graph correctly depicts the property mentioned against the indicated species.



6. In which of the following processes, the value of magnetic moment does not change ?



Space for Rough Work

SECTION-I (ii) : (Maximum Marks: 24)

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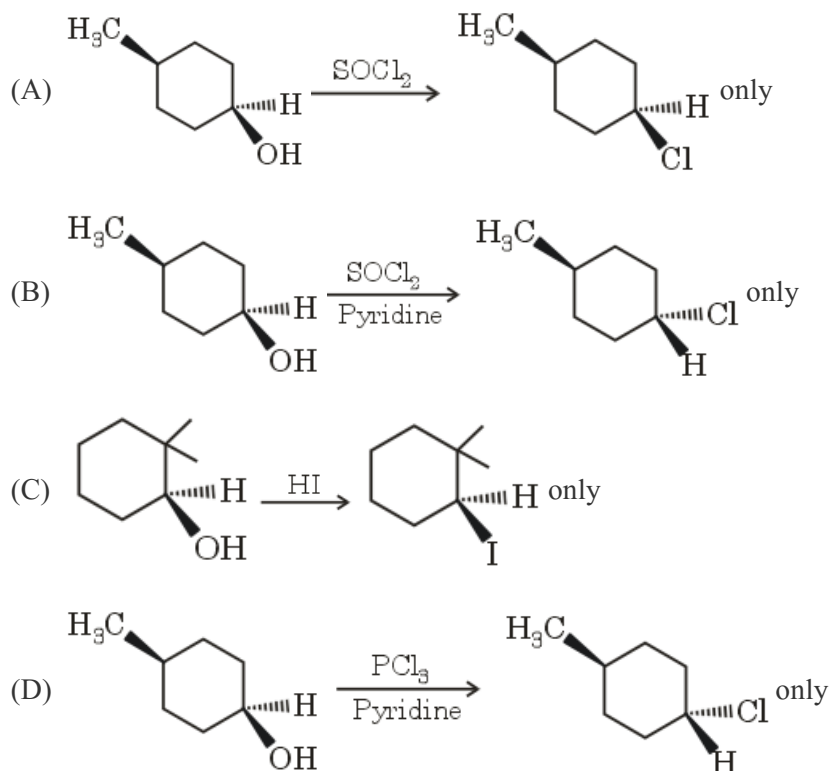
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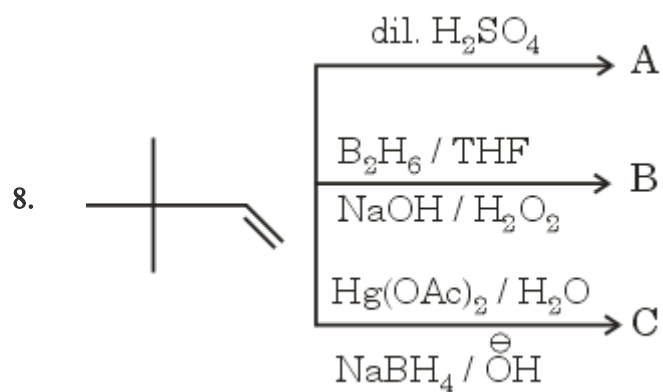
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7. Which of the following reaction is showing correct major product :





Select correct statement from the following about major product A, B and C :

- (A) A is 3° alcohol and show optical isomersim
- (B) B is 1° alcohol and does not show optical isomersim
- (C) C is 2° alcohol and shows optical isomersim
- (D) A and B are positional isomer of each other

Space for Rough Work

9. A metal 'M' crystallizes in two crystal systems fcc and hcp. If d -density, z - number of atoms per unit cell and a, b, c are edge lengths of the unit cell along the crystallographic axes then. Select the correct options :

(A) $\frac{d_{\text{fcc}}}{d_{\text{hcp}}} \neq 1$

(B) $\frac{a_{\text{fcc}}}{c_{\text{hcp}}} = \frac{\sqrt{3}}{2}$

(C) $\frac{d_{\text{fcc}}}{d_{\text{hcp}}} = 1$

(D) $\frac{z_{\text{fcc}}}{z_{\text{hcp}}} = \frac{2}{3}$

10. For an ideal binary liquid solution with $P_A^\circ > P_B^\circ$, which relation among the following is/are correct-

[Given : X_A = mole fraction of A in liquid phase

X_B = mole fraction of B in liquid phase]

Y_A = mole fraction of A in vapour phase

Y_B = mole fraction of B in vapour phase]

(A) $X_A < Y_A$

(B) $X_A > Y_A$

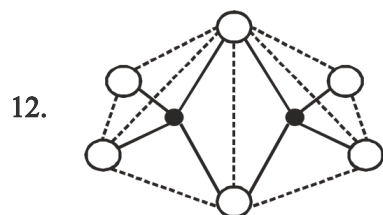
(C) $\frac{X_A}{X_B} < \frac{Y_A}{Y_B}$

(D) X_A and Y_A can't be co-related

Space for Rough Work

11. The compound AB_3 has more dipole moment than AC_3 . The compound AC_3 on action of superheated steam under high pressure yield an acidic oxide via S_N1 mechanism. AB_3 has tendency to unsymmetrically cleave bridge bond of B_2H_6 . Which is/are correct in view of above information.

- (A) The compound "BC" can be produced by hydrolysis of C_2 .
- (B) The compound BA_3 has less B–A bond length compared to C–A bond length in CA_3 .
- (C) AC_3 show back bonding.
- (D) A can form neutral oxide



above type of structure is observed in

- (A) $Si_2O_7^{6-}$
- (B) Al_2Cl_6
- (C) B_2H_6
- (D) I_2Cl_6

Space for Rough Work

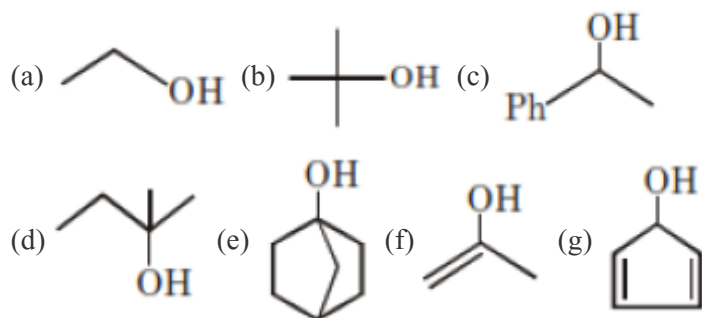
SECTION-II : (Maximum Marks: 24)

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- Answer to each question will be evaluated according to the following marking scheme:

Full Marks : +4 If ONLY the correct numerical value is entered.

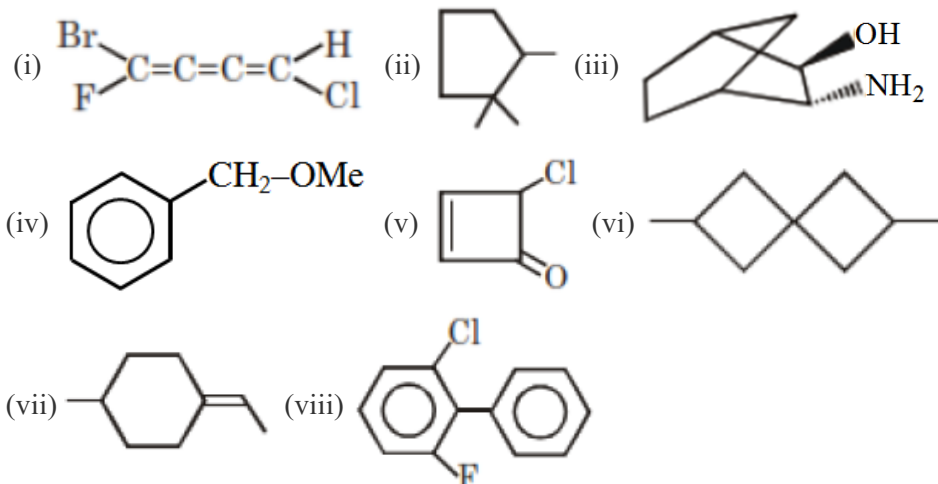
Zero Marks : 0 In all other cases.

1. How many of the following compound will form immediate turbidity with (HCl / ZnCl₂):



Space for Rough Work

2. Number of molecules show optical isomerism :

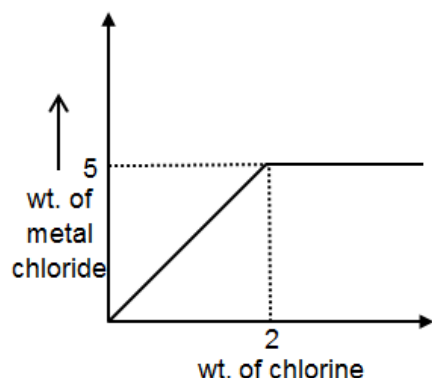


3. A compound has cubical unit cell in which 'A' atoms are found at six corners and 'B' atoms are present at remaining corners and at face centres which are not opposite to each other and 'C' atoms are present at body centres and remaining face centres. Calculate density of the compound in $\text{amu}/\text{\AA}^3$.

Given : A = 20 amu, B = 60 amu, C = 80 amu and edge length = $2\text{\AA}$.

Space for Rough Work

4. A metal chloride of empirical formula MCl is formed (where M is metal). In an experiment weight of metal chloride has following variation according to weight of chlorine.



Then atomic weight of metal (M) will be :

5. From the following find total number of molecule/ions which have bond angle is more than 109.28° is/are, among the following (w.r.t. underlined atom).
 $\underline{Si}H_4$; $\underline{Be}Cl_3^-$; $\underline{N}O_2$; $\underline{S}O_2$; $\underline{O}Cl_2$; $\underline{O}(CH_3)_2$; $\underline{O}(SiH_3)_2$
6. Find out the number of following orders which are **CORRECT** against the mentioned properties.
- (i) $I_2 > Br_2 > Cl_2 > F_2$: oxidising strength
 - (ii) $As^{+3} > Bi^{+3}$: reducing strength
 - (iii) $HgF_2 > HgCl_2 > HgBr_2 > HgI_2$: solubility in water
 - (iv) $NaI_3 > KI_3$: thermal stability
 - (v) $NaF < MgF_2 < AlF_3$: lattice energy
 - (vi) $KF > RbF > CsF > AgF$: Covalent character

Space for Rough Work

PART-3 : MATHEMATICS
SECTION-I (i) : (Maximum Marks: 18)

- This section contains **SIX (06)** questions.
- Each question has Four options (A), (B), (C) and (D). **ONLY ONE** of these four options is correct.
- For each question, choose the option corresponding to the correct answer.
- Answer to each question will be evaluated according to the following marking scheme :

Full Marks : +3 If **ONLY** the correct option is chosen.

Zero Marks : 0 If none of the options is chosen (i.e. the question is unanswered)

Negative Marks : -1 In all other cases

1. If $\tan^{-1}(x+1) + \cot^{-1}(x-1) = \sin^{-1} \frac{4}{5} + \cos^{-1} \frac{3}{5}$, then possible value of x is
 (A) $4\sqrt{\frac{3}{7}}$ (B) $4\sqrt{\frac{7}{3}}$ (C) $-4\sqrt{\frac{7}{3}}$ (D) $6\sqrt{7}$
2. Given that $f(x)$ is a function defined on $\mathbf{R}$, satisfying $f(1) = 1$ and for any $x \in \mathbf{R}$, $f(x+5) \geq f(x) + 5$ and $f(x+1) \leq f(x) + 1$. If $g(x) = f(x) + 5 - x$, then $g(2021)$ is equal to -
 (A) 3 (B) 4 (C) 5 (D) 6

Space for Rough Work

3. If $[x]^3 + [y]^3 + [z]^3 - 3[x][y][z] = 4$ where $x, y, z > 0$ then minimum possible value of $(x + y + z)$ is (where $[.]$ denotes greatest integer function)
- (A) 1 (B) 2 (C) 3 (D) 4
4. Let $(1 + \sqrt{2})^n = A_n + B_n\sqrt{2}$, where A_n and B_n are rational numbers and $n \in \mathbf{N}$, then the value of $\lim_{n \rightarrow \infty} \frac{A_n}{B_n}$ is equal to -
- (A) $\left[\lim_{x \rightarrow 0} \frac{\sin x}{x} \right]$
- (B) $\lim_{x \rightarrow 0} \left[\frac{\sin x}{x} \right]$
- (C) $\lim_{x \rightarrow \frac{\pi}{4}} \frac{|\cos x - \sin x|}{\left| x - \frac{\pi}{4} \right|}$
- (D) $\left[\lim_{x \rightarrow 0} \frac{\sin \sqrt{2} x}{x} \right]$
- (where $[.]$ denotes greatest integer function)

Space for Rough Work

5. $\frac{\sum_{n=1}^{99} \sqrt{10 + \sqrt{n}}}{\sum_{n=1}^{99} \sqrt{10 - \sqrt{n}}}$ equals

(A) $\sqrt{2}$

(B) $\sqrt{2} - 1$

(C) $\sqrt{2} + 1$

(D) None of these

6. Value of $\lim_{x \rightarrow 1} \frac{(\log(1+x) - \log 2)(3 \cdot 4^{x-1} - 3x)}{((7+x)^{1/3} - (1+3x)^{1/2}) \sin \pi x}$ is equal to

(A) $\frac{9}{\pi} (2 \log 2 - 1)$

(B) $\frac{9}{4\pi} [\log 4 - 1]$

(C) $\frac{9}{2\pi} \left[\log 4 - \frac{1}{2} \right]$

(D) $\frac{2}{3\pi} [2 \log 2 - 1]$

Space for Rough Work

SECTION-I (ii) : (Maximum Marks: 24)

- This section contains **SIX (06)** questions.
- Each question has **FOUR** options. **ONE OR MORE THAN ONE** of these four option(s) is (are) correct answer(s).
- For each question, choose the option(s) corresponding to (all) the correct answer(s)
- Answer to each question will be evaluated according to the following marking scheme:

Full Marks : +4 If only (all) the correct option(s) is (are) chosen.

Partial Marks : +3 If all the four options are correct but **ONLY** three options are chosen.

Partial Marks : +2 If three or more options are correct but **ONLY** two options are chosen and both of which are correct.

Partial Marks : +1 If two or more options are correct but **ONLY** one option is chosen and it is a correct option.

Zero Marks : 0 If none of the options is chosen (i.e. the question is unanswered).

Negative Marks : -2 In all other cases.

- **For Example** : If first, third and fourth are the **ONLY** three correct options for a question with second option being an incorrect option; selecting only all the three correct options will result in +4 marks. Selecting only two of the three correct options (e.g. the first and fourth options), without selecting any incorrect option (second option in this case), will result in +2 marks. Selecting only one of the three correct options (either first or third or fourth option), without selecting any incorrect option (second option in this case), will result in +1 marks. Selecting any incorrect option(s) (second option in this case), with or without selection of any correct option(s) will result in -2 marks.

7. Given $\lim_{x \rightarrow \alpha} \frac{\sin([x^3] - [x]^3)}{(x - \alpha)^3}$ exists finitely, then the value of α can be (where $\alpha \in I$ & $[.]$ denotes greatest integer function) -
- (A) 0 (B) 1 (C) -1 (D) 2

Space for Rough Work

8. Let $f(x) = \begin{cases} \min \{ \sin^{-1} x, \cos^{-1} x \} & ; \quad 0 \leq x \leq 1 \\ \max \{ \sin^{-1} x, \cos^{-1} x \} & ; \quad -1 \leq x < 0 \end{cases}$ and $f(x) = \frac{1}{2}$, $f(x) = \frac{1}{4}$ & $f(x) = 2$ has number of solution(s) $\lambda_1, \lambda_2, \lambda_3$ respectively, then

(A) $\lambda_1 = 2$ (B) $\lambda_2 = 1$

(C) $\lambda_2 = 2$ (D) $\lambda_3 = 1$

9. A right angled triangle has one leg of length 1, another of length 'x' and hypotenuse of length 'y'. The angle opposite to the side x is θ then -

(A) $y^n - x^n = \frac{1 - \sin^n \theta}{\cos^n \theta}$

(B) $\lim_{\theta \rightarrow \frac{\pi}{2}} (y - x) = 0$

(C) $\lim_{\theta \rightarrow \frac{\pi}{2}} (y^2 - x^2) = 1$

(D) $\lim_{\theta \rightarrow \frac{\pi}{2}} (y^3 - x^3)$ does not exist

Space for Rough Work

10. If a, b, c are positive real numbers such that $\frac{1}{1+a^4} + \frac{1}{1+b^4} + \frac{1}{1+c^4} = \frac{\sqrt{1+a^4} + \sqrt{1+b^4} + \sqrt{1+c^4}}{\sqrt{(1+a^4)(1+b^4)(1+c^4)}}$, then which of the following statement(s) is/are true ?

(A) $\frac{a}{b} + \frac{b}{c} + \frac{c}{a} \leq 3$

(B) $\frac{a}{b} + \frac{b}{c} + \frac{c}{a} \geq 3$

(C) $(a+b+c) \left(\frac{1}{a} + \frac{1}{b} + \frac{1}{c} \right) \leq 9$

(D) $(a+b+c) \left(\frac{1}{a} + \frac{1}{b} + \frac{1}{c} \right) \geq 9$

Space for Rough Work

11. The value of $\sum_{r=2}^{100} \frac{2^r (2-r)}{r(r+1)(r+2)}$ is equal to

(A) $\frac{2}{3} - \frac{2^{100}}{(51)(101)}$

(B) $\frac{2}{3} - \frac{2^{101}}{(101)(102)}$

(C) $\frac{1}{2} - \frac{2^{100}}{100.101}$

(D) $\frac{1}{2} - \frac{2^{100}}{(101)(51)}$

12. For the equation $9^{\log_3(\log_2 x)} = \log_2 x - \log_2^2 x + 1$ is -

(A) number of solutions is 1

(B) number of solutions is 2

(C) sum of all the solutions is prime number

(D) sum of the solutions is $2 + \sqrt{3}$

Space for Rough Work

SECTION-II : (Maximum Marks: 24)

- This section contains **SIX (06)** questions. The answer to each question is a **NUMERICAL VALUE**.
- For each question, enter the correct numerical value of the answer in the place designated to enter the answer. If the numerical value has more than two decimal places, **truncate/round-off** the value to **Two** decimal places; e.g. 6.25, 7.00, -0.33, -.30, 30.27, -127.30, if answer is 11.36777..... then both 11.36 and 11.37 will be correct)
- Answer to each question will be evaluated according to the following marking scheme:

Full Marks : +4 If ONLY the correct numerical value is entered.

Zero Marks : 0 In all other cases.

-
1. Let $A = \{1, 2, 3, 4\}$, $B = \{1, 2, 3, 4\}$. The total number of functions $f : A \rightarrow B$ such that $f(x) \neq x \forall x \in A$ and range of $f(x)$ contains exactly three elements, is
 2. $\lim_{x \rightarrow -\infty} \left(\sqrt{x^2 - \frac{4x}{3}} + x + 2 \right)$ is equal to -
-

Space for Rough Work

3. If $\sum_{n=0}^{\infty} \arccot \left(\frac{n^2 + n + 4}{2} \right) = k\pi$, then value of k is equal to
4. A function $f : A \rightarrow B$ is defined such that set A consists of 6 different elements and set B consists of 4 different elements, then number of onto functions f is
5. Let $f(x) = a_0 + a_1x + a_2x^2 + a_3x^3 + a_4x^4$ where $a_i \in \mathbb{R}$ and $\lim_{x \rightarrow 0} \left(\frac{f(x) - x + 2x^2}{3x^3} \right)^{\frac{2}{x}} = e^4$, then $\sum_{i=0}^4 a_i$ is equal to
6. Number of values of x in $[0, \pi]$ for which $\sin^{-1}(\sin 6x) = \cos^{-1}(\cos x)$ is -

Space for Rough Work

Space for Rough Work

NAME OF THE CANDIDATE

FORM NO.

I have read all the instructions
and shall abide by them.

Signature of the Candidate

I have verified the identity, name and Form
number of the candidate, and that question
paper and ORS codes are the same.

Signature of the Invigilator

Space for Rough Work